# Analyzing College Students Awareness and Preparedness Toward Flood Risks in the Philippines: A Data Science Approach

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**Introduction**

Flooding is one of the recurring hazards experienced in different parts of the Philippines, particularly during periods of heavy rainfall, typhoons, and other weather-related events. It can affect communities by damaging homes and infrastructure, disrupting transportation and education, and putting people's safety at risk. Because students are also members of communities that may be exposed to flooding, understanding their level of awareness and preparedness is important.

This study aims to **analyze the awareness and preparedness of college students regarding flood risks in the Philippines**. It will examine how knowledgeable students are about the causes, warning signs, risks, and safety measures associated with flooding. It will also determine how prepared they are to respond if a flood occurs in their community, including their knowledge of evacuation procedures, emergency supplies, evacuation areas, emergency contacts, and actions to take before, during, and after a flood.

The study will gather data through a **survey questionnaire administered to college students**. The collected data will be processed and analyzed using **Data Science techniques**, such as data cleaning, descriptive statistics, data visualization, correlation analysis, and potentially clustering. These methods can be used to identify patterns in students' awareness and preparedness and determine whether factors such as previous flood experience, location, access to disaster information, and participation in disaster preparedness activities are associated with their level of preparedness.

The findings may help identify **gaps in students' knowledge and preparedness** and provide useful information for schools and communities in developing better flood-awareness and disaster-preparedness programs.

This study looks at how aware college students in the Philippines about flood risks and how ready they are to handle such disasters. It focuses on understanding the connection between where students live whether with their families or in boarding houses and dorms and their past experiences with floods. The goal is to see how these factors affect their level of preparedness. Using data-driven methods the study aims to uncover where the biggest gaps are in disaster readiness, among students.

To achieve these objectives, the study aims to answer the following specific research questions:

1. What is the demographic and situational profile of the respondents in terms of:

· Age, Sex, Year Level, and Course/Program?

· Living Arrangement (Living with family/parents, Boarding house/dorm/rented room, or Living with relatives)?

· Municipality of residence?

· History of flood experience in their current residence?

2. What is the level of flood risk awareness among college students regarding:

· Weather advisories and PAGASA color-coded warning signals?

· Local flood history and risk level of their current area?

· Emergency contact numbers of local disaster response units (e.g., MDRRMO/PDRRMO, Red Cross, BFP)?

· Locations of designated evacuation centers and safe routes?

· Health risks and waterborne diseases associated with floodwaters?

3. What is the level of personal flood preparedness among college students in terms of:

· Maintaining a packed emergency "Go-Bag"?

· Storing non-perishable food items and clean drinking water?

· Having a clear emergency communication and evacuation plan?

· Securing important personal documents in waterproof storage?

· Actively monitoring official weather advisories?

· Knowing electrical safety protocols during heavy flooding?

4. Is there a significant difference in the levels of flood risk awareness and preparedness when respondents are grouped according to their living arrangement and flood history?

5. Using data science techniques, what key patterns, preparedness gaps, and determinant factors emerge to inform community and institutional flood safety recommendations?

Survey Link:
https://docs.google.com/forms/d/e/1FAIpQLSdUhsm9eaOW3yiY22-PEcXoh-lzWM7G-AjXeKrStOKrNPXHCQ/viewform?usp=dialog